

PRD: AI Powered Search Campaign Bid Optimization - Instacart POC

Capability Domain	Marketing Automation
Capability Module	Pacing and Forecast AI Recommendation and Application

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Goal

Transform the current manual and reactive management process for search campaigns/ad groups into a proactive, data-driven system supported by ML/AI.

This capability is a key component of the broader **Smart Media Investment Initiative**, which also includes Sales Driver, Predictive Budget Planning, Campaign Optimizer, and Share Driver models.

Customer Segment	Unique Value Proposition	Measure of Success
- Search Marketing Leads - Search Ad Operations This capability primarily benefits Search Marketing	This system will empower Search Marketing managers and leads to make optimized spend decisions within search	Performance of AI recommended ad group traffic statuses and bids, and key word bids

Managers and Leads who are directly responsible for search campaign performance and real time bid values.

platforms, predict sales outcomes driven by search, and respond dynamically to market changes in the search landscape. Leveraging in-house ML models, first and third party data, and Generative AI, this system will provide actionable insights and automation at the status and bid level for ad groups and keywords within campaigns.

Customer Need

The current search forecast management process is largely manual, reactive, and lacks integrated, real-time intelligence for optimal decision-making. Key challenges include:

- **Inaccurate & Reactive Projections:** Relying heavily on historical data and manual adjustments leads to forecasts that often diverge significantly from actual performance, impacting downstream financial planning. Responding to real-time market shifts (e.g., competitive surges, unexpected budget changes, seasonality) requires time-consuming manual adjustments, leading to missed opportunities or sub-optimal spend.
- **Sub-optimal & Opaque Line Item Spend:** Bids and traffic allocation for ad groups and key words across brands, periods, and campaign types is often based on intuition or static rules, missing opportunities to maximize sales, incrementality, or other objectives. The rationale behind bid adjustments is seldom clear, hindering trust and preventing a deeper understanding of key performance drivers.
- **Limited Insight into Detailed Performance drivers:** Real time bid optimization struggles to accurately quantify and provide quick responses to the impact of bid values on sales and share, making it difficult to justify the overall budget asks or assess risk.

Existing Alternative

Currently only available for limited Instacart brands as an Excel table that has to be manually implemented in Skai

Product Construct

The primary goals of the Search Bid Optimization system are to:

- **For Weekly Management:** Improve Search Forecast Accuracy and Optimize Search Bud Efficiency for immediate, actionable insights and outcomes.
- **For Strategic Planning:** Enhance Search Decision-Making Agility and Quantify Search Risk and Opportunity for long-term budget effectiveness.
- **Overall:** Foster collaboration between Search Marketing and day to day ad ops.

Typical Customer Journeys

Campaign Optimization: Weekly Predictive Insights & Bid Optimization

- As a **Search Marketing Manager**, I need to receive, **weekly optimized bid and status recommendations at the campaign level** that compare an 'AI Spend Recommendation' to my 'Current Bids' and show the predicted sales impact for both, so I can confidently adjust active state and bid values at both line item and key word levels.
- This allows me to maximize attributed sales within my specified budget constraints at a granular campaign level.

Capabilities

Provide AI recommended ad group status

- Provide recommended pause/unpause status at the ad group level and resulting outcome (Sector/BU/Category/Brand)
- Provide recommended pause/unpause status at the ad group level and resulting outcome by metric (RSV/Incremental Sales/ROI/Share)

Provide AI recommended ad group bid values

- Provide recommended bid values at the ad group level and resulting outcome (Sector/BU/Category/Brand)
- Provide recommended bid values at the ad group level and resulting outcome by metric (RSV/Incremental Sales/ROI/Share)

Provide AI recommended key word bid values

- Provide recommended bid values at the key word level and resulting outcome (Sector/BU/Category/Brand)
- Provide recommended bid values at the key word level and resulting outcome by metric (RSV/Incremental Sales/ROI/Share)

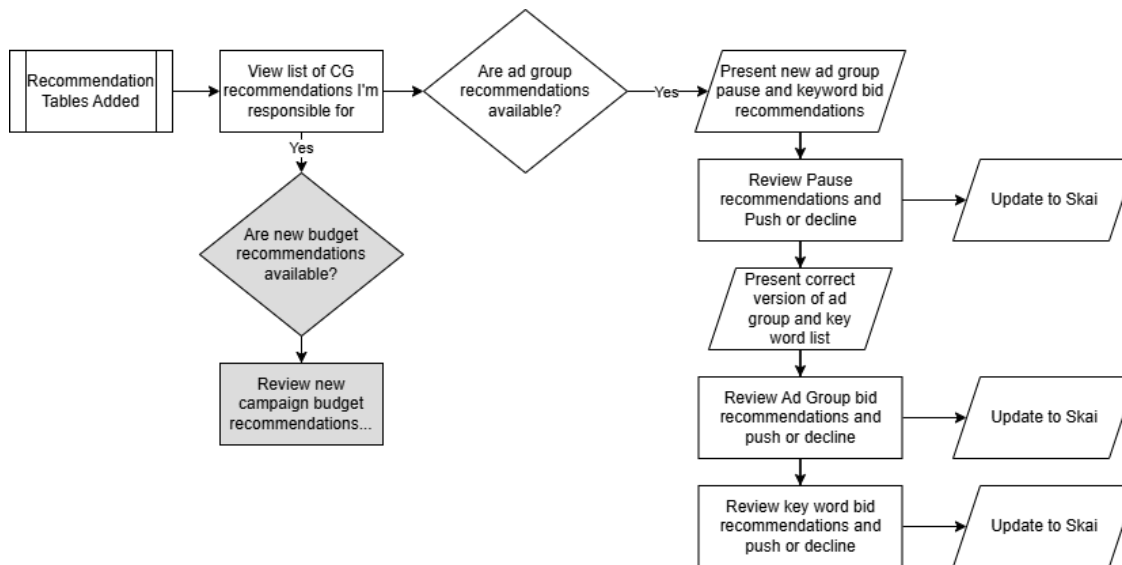
Present forecast of recommended changes to bids and ad group status

- Update forecast charts shows projected outcome of status and bid changes
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Functional Requirements

See also [AUTO-8046: Instacart Weekly Ad Group Recommendation POC](#) BACKLOG

User/Platform Workflow



Campaign Optimization Recommendation Tables

Weekly optimization recommendations need to be created for specific customers, for every BU. For POC this is just for one Customer Group: **Instacart - US - Total Instacart**, and therefore will only be visible to anyone who has that scoped authorization.

This consists of three separate table, but there needs to be interactions among the table to account for status

- Ad Group Pause status and Bid recommendations: [Campaign Optimizer \(Bid Optimization\) Recommendation Action | galileo_marketing_recommendation_ad_group_actions](#)
- Key Word Bid values, current and recommended: [Campaign Optimizer \(Bid Optimization\) Recommendation Action | galileo_marketing_recommendation_keyword_actions](#)
- Sales projected outcomes: [Campaign Optimizer \(Bid Optimization\) Recommendation Action | galileo_marketing_recommendation_sales_prediction](#)
 - Used for both table and chart

See [Campaign Optimizer \(Bid Optimization\) Recommendation Action](#) for other data model details

Snowflake Tables Specs and Names:

- [galileo_marketing_recommendation](#)
 - Might evolve or be included in the recommendation object that needs to be created for [PRD: Recommendation Object Management](#)

- [galileo_marketing_recommendation_ad_group_actions](#)
- [galileo_marketing_recommendation_keyword_actions](#)
- [galileo_marketing_recommendation_sales_prediction](#)

The schema is `PEP_PROD_AZURE_BUSINESS_VAULT.INFO MART_BUSINESS_VAULT`

Original ML Data

This is the current Excel that is being delivered to the Instacart search marketing team, included for reference



Future Directions

- Layer in finance taxonomy for consistency, if possible
 - Currently no reliable way to link these with campaigns, but possibly we can with ad groups?
 - Need to explore if this will be necessary
- Create permutations of bid rows with and without all paused from pause recommendation table, or with only AI paused, so paused rows are excluded from
- For ad group bids show only ad groups that do not have all manual overrides at the keyword level
- The tables should include any possible AI confidence settings, once these are available
- Re-compute sales projections from updated bid values and pause settings
- Auto-refresh with any recent budget updates prior to expiration

New: Recommendations List

See [PRD: Recommendation Object Management](#)

- The campaign ad group recommendations will be viewed and opened from the Recommendations List

Campaign Optimization Page

Should be a page and not a modal.

This is the ability to open and review all campaign line item status and bid recommendations. This view requires:

Selections of Optimization Metric

- Currently only RSV, so we don't need these cards yet for the POC
- Later should follow the design pattern for multiple metrics as used for budget optimization

Projected Optimized vs Non-optimized Campaign Sales Chart

There should be a simple chart comparing the optimized vs non-optimized outcomes of sales, showing projected, baseline*, and actual.

Uses data from this table: [galileo_marketing_recommendation_sales_prediction](#)

This uses the data currently available from the sample

*Note: For consistency this will need implementation of continued baseline projected values from the ML team.

Actionable Recommendations

Consists of Pausing, Ad Group Default Bids, and Individual Keyword Bid recommendations.

- Each step has a separate table of values and a global acceptance control. See below

Review and Set Recommended Pause Status

This step presents a list of the ad groups that should be paused, sorted by campaign. Users can choose to select which ad group to pause.

Uses data from both [galileo_marketing_recommendation_ad_group_actions](#) and [galileo_marketing_recommendation_sales_prediction](#)

Accepting flags all selected ad groups for pausing.

Common Table Fields

- Select*
 - *For POC we should strongly consider skipping any partial acceptance as we currently cannot implement a sales forecast update on only a partial recommendation application
- Campaign
 - Name of campaign in Skai
 - Link to Skai campaign using the URL form:
`https://pepsico.kenshoo.com/grid.jsp#/analysis/adgroup?campaignId=[campaign id]`
 - Currently Skai campaign ID is not returned within ML so would required some lookup
- Ad Group
 - Name of ad group within campaign in Skai
 - Only ad groups marked as PAUSE for AD_GROUP_STATE are shown
 - Pause occurs at the ad group level

- Link to Skai campaign using the URL form:

`https://pepsico.kenshoo.com/grid.jsp#/analysis/keyword?adGroupId=[ad group id]&[campaign id]`

- Currently Skai ad group ID is not returned within ML so would required some lookup

Campaign Metrics

Need to clearly indicate these metrics are at the campaign level, not the ad group level

- Optimized Spend
- Spend
 - Projected for week
 - Replaced with actual for historical
- Optimized Attributed Sales
- Attributed Sales
 - Projected for week
 - Replaced with actual for historical
- Optimized Total Sales
- Total Sales
 - Projected for week
 - Replaced with actual for historical
- Optimized ROAS
 - `Optimized Attributed Sales - Optimized Spend`
- Projected ROAS
 - `Attributed Sales - Spend`
 - Projected for week
 - Replaced with actual for historical

Review and Set Recommended Ad Group Bids

Only ad groups marked as `ACTIVE` for `AD_GROUP_STATE` are shown

Same as Pause table, except that these fields are added after the Ad Group fields:

- Current Bid
- Optimized Bid

Review and Set Recommended Search Key Word Override Bids

Uses [galileo_marketing_recommendation_keyword_actions](#) table

Same as Pause table, except that these fields are added after the Ad Group fields:

- Key Word
- Current Bid
- Optimized Bid

Complete and Return to Budget Optimization

Before applying all changes review updates and then accept to push to Skai, or reject

- Count and List Paused Ad Groups with net spend decrease
- Count of Updated Ad Group bids broken down by increase/decrease and net ROAS
- Count of Updated Key Word bids broken down by increase/decrease and net ROAS

UX Requirements

WIP  Forecast

Appendix

 [Campaign Optimizer \(Bid Optimization\) Recommendation Action](#)

 [\[Galileo\]\[Design Doc\] Bid Optimization Recommendation Action](#)

In progress design:  Forecast

Jira story:  [AUTO-8046: Instacart Weekly Ad Group Recommendation POC](#) 